

Project Name: Cognitive Canvas

1.0 The Problem

Journaling is a powerful tool for mental health, but traditional methods are static. We write entries in isolation, making it nearly impossible to "see the forest for the trees." Users struggle to identify recurring behavioral patterns, spot cognitive distortions, or understand the root causes connecting their bad days. Valuable insights remain buried in pages of text.

2.0 The Solution

Cognitive Canvas transforms journaling from a passive record into an active, intelligent model of your mind. It is a full-stack application that uses AI to analyze daily reflections and visualize them as an evolving, interactive network. Instead of just reading past entries, users can explore a living map of their psychology.

3.0 Key Features

- **AI-Powered Analysis:** The system automatically detects emotions, identifies 12 types of cognitive distortions (e.g., catastrophizing, black-and-white thinking), and extracts causal chains (Trigger → Thought → Emotion).
- **Interactive Visualization:** A D3.js force-directed graph represents the user's mental landscape. Entries, emotions, and themes appear as nodes that cluster together physically based on context, revealing hidden connections.
- **Emotional Weather Map:** Going beyond simple mood tracking, this feature calculates "emotional volatility" and maps transition patterns, showing how specific emotions cascade into others over time.
- **Guided Root Cause Analysis:** An interactive chat mode that uses Socratic questioning to help users trace surface-level emotions down to core beliefs and unmet needs.

4.0 How We Use Claude

Claude 3.5 Sonnet is the cognitive engine behind the entire platform. We leverage its superior reasoning and nuance in two specific ways:

1. **Structured Psychological Analysis:** We use Claude to parse unstructured journal text into complex JSON schemas. It identifies subtle causal links between events and emotions that keyword-based systems miss.
2. **Compassionate Reasoning:** In our "Root Cause" sessions, Claude is prompted to act as a therapeutic guide using the "5 Whys" technique. It maintains context across turns to ask probing, empathetic questions that guide the user to their own realizations, rather than just offering generic advice.

5.0 Impact

Cognitive Canvas empowers users to build genuine self-awareness. By making abstract thought patterns visible and tangible, we help users move from passive venting to active growth. Built with a privacy-first architecture (local storage options and no third-party tracking), it provides a safe, intelligent space for deep self-reflection.